

```

import pandas as pd
from sklearn.naive_bayes import GaussianNB
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

data = pd.read_csv("pima_indian.csv")
Data

# Step 2: Separate features and target
# Assuming last column is Outcome (0 or 1)
X = data.iloc[:, :-1] # Features
y = data.iloc[:, -1]  # Target

# Step 3: Split dataset into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=0
)

# Step 4: Train Naive Bayes model
model = GaussianNB()
model.fit(X_train, y_train)

# Step 5: Predict on test data
y_pred = model.predict(X_test)

# Step 6: Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Predicted values:", y_pred)
print("Actual values:", y_test.values)
print("Accuracy of Naive Bayes Classifier:", accuracy * 100, "%")

```

```

import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from sklearn.naive_bayes import GaussianNB

data
data.isnull().sum()

```

```
data.duplicated()
```

```
data.replace('?',pd.NA,inplace=True)
```

```
data.dropna(inplace=True)
```

```
X=data[['age','sex','cp','trestbps','chol','fbs','restecg','thalach','exang','oldpeak','slope','ca','thal']].a  
stype(float)
```

```
y=data['heartdisease']
```

```
y=(y>0).astype(int)
```

```
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
```

```
model=GaussianNB()
```

```
model.fit(X_train,y_train)
```

```
y_pred=model.predict(X_test)
```

```
print(accuracy_score(y_test,y_pred))
```

```
import pandas as pd
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.metrics import accuracy_score,precision_score,recall_score
```

```
from sklearn.naive_bayes import GaussianNB
```

```
data=pd.read_csv("titanic.csv")
```

```
data
```

```
data.replace("?",pd.NA,inplace=True)
```

```
data.dropna(inplace=True)
```

```
X = data[['Pclass','Sex','Age','SibSp','Parch','Fare','Embarked']]
```

```
y=data['Survived']
```

```
X['Sex'] = X['Sex'].map({'male':0, 'female':1})
```

```
X['Embarked'] = X['Embarked'].map({'C':0, 'Q':1, 'S':2})
```

```
X_train, X_test, y_train, y_test = train_test_split(
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```
    X, y, test_size=0.2, random_state=0
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```
)
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```
# Step 4: Train Naive Bayes model
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print("Accuracy of Naive Bayes Classifier:", accuracy * 100, "%")
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import numpy as np
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```

```
data=pd.read_csv('winequalityN.csv')
data.isnull().sum()
data.dropna(inplace=True)
data.duplicated().sum()
data.drop_duplicates(inplace=True)
```

```
data['type'].value_counts()
data['type']=data['type'].map({'white':0,'red':1})
data
```

```
X=data[['fixed acidity','volatile acidity','citric acid','residual sugar','chlorides','free sulfur
dioxide','total sulfur dioxide','density','pH','sulphates','alcohol','quality']]
y=data['type']
```

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```
from sklearn.naive_bayes import GaussianNB

data=pd.read_csv('customer_churn.csv')

data.isnull().sum()
data.duplicated().sum()

data['Gender']=data['Gender'].map({'Female':0,'Male':1})
data['Subscription Type']=data['Subscription Type'].map({'Basic':0,'Standard':1,'Premium':2})
data['Contract Length']=data['Contract Length'].map({'Monthly':0,'Quarterly':1,'Annual':2})

X=data[['Age','Gender','Tenure','Usage Frequency','Support Calls','Payment Delay','Subscription
Type','Contract Length','Total Spend','Last Interaction']]
y=data['Churn']

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
model=GaussianNB()
model.fit(X_train,y_train)
y_pred=model.predict(X_test)
print(accuracy_score(y_test,y_pred))
```